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MILITARY SPECIFICATION

HOWITZER, MEDIUM, SELF-PROPELLED: M109 (FRG)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one type of full-tracked, self-propelled vehicle to be supplied under a Federal Republic of Germany contract.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

SPECIFICATIONS

Federal

VV-F-800	- Fuel Oil, Diesel.

Military

MIL-P-514	- Plates, Identification, Instruction and Marking, Blank.
MIL-M-45212	- Mount, Gun; Mount, Howitzer.
MIL-A-62048	- Air Cleaners, Automotive, Heavy Duty, Dry Type (for Internal Combustion Engines).

STANDARDS

Federal

FED. STD. No. 595	- Colors.

Military

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.

FSC-2350

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MIL-STD-193	- Painting Procedures Tactical Vehicle (Tracked and Wheeled).
MIL-STD-417	- Rubber compositions, Vulcanized General Purpose, Solid (Symbols & Tests).
MIL-STD-461	- Electromagnetic Interference Characteristics Requirements for Equipments
MIL-STD-862	- System Safety Program for System and Associated Sub-Systems and Equipment, Requirements for.

PUBLICATIONS

Engineering Parts List

10941625	- Howitzer, Medium, Self-Propelled: M109 (FRG).

Supplemental Quality Assurance Provisions (SQAP)

10941625	- Howitzer, Medium, Self-Propelled: M109 (FRG).

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Preproduction model. When specified (see 6.2), a preproduction model(s) of the vehicle to be furnished under this specification shall be produced prior to the manufacture or fabrication of vehicles in quantity. The vehicle(s) when complete shall be submitted to the Government for examination and tests to determine conformance to the requirements of this specification. Vehicle(s) submitted by the contractor shall be fully representative of vehicles proposed to be supplied by the vehicle manufacturer from production facilities and tooling under the supply contract.

3.2 Initial production vehicles. The vehicles selected shall be fully representative of vehicles proposed to be furnished under the contract, shall be submitted to the Government for inspection to determine conformance to the requirements of this specification.

3.2.1 Reliability (automotive). The vehicle(s) mean miles between failure (MMBF) shall not be less than 285 miles during the first 4000 miles of operation consisting of 25 percent hard surface roads (HSR), 50 percent secondary roads (SR), and 25 percent cross country (CC) operation. A failure for calculation of Mean Miles Between Failure (MMBF) shall be as defined by the AMC/TRADOC Artillery Failure Definition, dated 13 December 1972. The definition reads (excluding the amplification paragraphs): For the purpose of assessing reliability, a failure is defined as. any malfunction which the operator/crew cannot or is not authorized to remedy by adjustment, repair, or replacement action within five minutes, using controls, Basic Issue Items (BII), Items Troop Installed or Authorized (OTIA) and parts authorized to the crew, and which causes or could cause inability to commence or the cessation of any mode of operation; a critical or catastrophic hazard to personnel or equipment as defined by MIL-STD-882 (15 June 1969). For related malfunctions only, the primary malfunction will be counted against reliability.

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3.2.2 Durability (Automotive). The vehicle, with rated payload shall have a .50 probability of completing the first 400 operation hours without replacement, rebuild or overhaul of powertrain components (Engine, Transmission and Final Drive).

3.2.3 Maintainability (Automotive). The total scheduled maintenance excluding driver/crew checks and services shall not exceed 125 manhours and the total unscheduled maintenance shall not exceed 80 manhours during 4000 miles operation. The scheduled maintenance intervals shall not be less than go days or 750 miles which ever comes first. Maximum time to repair (to include diagnosis repair and verification) utilizing personnel normally employed at:

(1) Organization maintenance shall not exceed 6 hours 95 percent of the time.
(2) Direct support maintenance shall not exceed 12 hours 95 percent of the time.
(3) General support maintenance shall not exceed 18 hours 95 percent of the time in a backup direct support role.

3.3 Materials.

3.3.1 Qualified products. Vehicle manufacturers shall be responsible for ascertaining that all assemblies and component parts entering into the construction of this vehicle, which require product qualification in accordance with a specification, are listed on respective qualified products lists (QPL) or have been approved for inclusion on such list. Manufacturer's inspection record shall specifically list such components, name of supplier(s), and number and date of applicable QPL.

3.3.2 Ozone resistance. Unless otherwise specified on the drawing, rubber components which are completely defined on Ordnance drawings and are flexed in any manner and are exposed to the atmosphere shall be ozone resistant to the degree that is required for the material to meet "C_1" test requirements of Standard MIL-STD-417.

3.4 Construction. Vehicles, components, sub-assemblies, assemblies, shall be fabricated and assembled into a vehicle in accordance with drawings listed or referred to in Engineering Parts Lists (EPL) 10941625. All parts requiring piece markings shall be piece marked as specified on drawings. All stenciling of vehicle pertaining to stowage shall be done per installation, stowage, Drawing 10941745 and 109410898.

3.4.1 Sealing and seals. After fording or floating for 15 minutes, the water contamination content of the engine, transmission, final drives, and suspension system lubricants shall be not more than 2 percent (by volume).

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3.4.1.1 Sealer. The sealer used shall restrict the entrance of water into the hull when vehicle is fording streams of rivers during cross-country operation.

3.4.1.2 Seals. Seals shall restrict the entrance of all foreign matter into the bearings when fording shallow water or operating in deep water. All seals shall prevent the leaking of lubricants from the bearings.

3.4.1.3 Hatch seals. With hatch closed, the seals shall prevent the entrance of water into the occupied portions of the vehicle.

3.4.2 Special kits. When specified (see 6.2), special kits shall be furnished as follows:

3.4.2.1 Winterization kit. With the winterization kit installed, the vehicle shall perform satisfactory when vehicle is operating in subzero ambient air temperatures down to minus 65°F.

3.4.3 Carbon monoxide concentration. The exhaust system shall prevent the accumulation of carbon monoxide in the occupied portions of the vehicle in excess of 0.01 percent under normal conditions of vehicle operation.

3.4.4 Controls. Controls for the engine, transmission, electrical equipment, permanently mounted fire extinguisher system, armament and other operating mechanisms shall operate without Interference.

3.4.5 Throttle linkage. With throttle linkage attached to engine, and throttle pedal fully depressed, the throttle shall be full rack position. With the throttle at full rack position, the throttle pedal shall be In contact with pedal stop.

3.4.6 Adjustment. All adjusting mechanisms shall operate without interference, and shall maintain adjustment settings under all operating conditions.

3.4.7 Fasteners and locks. All fastening and locking devices shall be positive locking. Means for locking vehicle from outside shall be provided.

3.4.8 Fuel system. During engine operation, the fuel system shall maintain fuel supply to The engine when ascending and descending 60 percent grades in forward and reverse gear, and when vehicle is being operated on 40 percent side slopes, with each side of vehicle up slope.

3.4.8.1 Fuel tanks. Provisions shall be made to assure the internal cleanliness of the fuel tanks prior to initial fueling of vehicle. The fuel tanks shall be capable of receiving fuel at a rate of 50 gallons per minute (gpm), and shall be free from leakage after Installation and vehicle break-in.

3.4.8.2 Fuel lines. Provisions shall be made to assure the internal cleanliness of fuel lines and connections prior to final assembly or initial fueling of vehicle. All fuel lines and connections shall be free from leakage.

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3.4.9 Hydraulic reservoir and lines. Provisions shall be made to assure the internal cleanliness of the hydraulic reservoirs lines, and connections prior to initial filling of the reservoir. The system shall be free from leakage, under all conditions of vehicle operation.

3.4.10 Cooling system. The engine water temperature, engine lubricant and transmission lubricant temperatures, shall not exceed those shown in table I when the vehicle in operated under any one of a combination of the following conditions:

(a) Ambient air temperature up to 115°F.
(b) Under full load.
(c) At all speeds down to the track slippage point of the vehicle when operating on smooth, level, hard-surfaced roads, but not when less than 3 miles per hour (mph) in first gear or below a transmission speed ratio of 0.56 in other gears.

Table I. Operating Temperatures

Water Temp. from engine	Maximum engine oil temp. in sump	Maximum transmission oil temp. into cooler
230°F	275°F	300°F

3.4.11 Electrical system.

3.4.11.1 Lights. The internal and external lights as specified on applicable drawings shall operate throughout all vehicle operating conditions.

3.4.11.2 Bilge pump. The system shall provide nominal 24 volts dc to the pump at all times when the impeller is covered by water.

3.4.11.3 Air cleaner blower. The air cleaner motor shall be capable of providing a flow of air through the blower outlet.

3.4.11.4 Personnel heater. The system shall provide sufficient voltage to operate the igniter and blower motor.

3.4.11.5 Communication equipment electrical service. The communication equipment electrical service shall be capable of delivering 24 volts ac nominal to the communication equipment.

3.4.11.6 Slip ring. The slip ring and external communications circuits shall maintain continuity without evidence of shorts.

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3.4.12 Repair welding. Repair welding of any type or class shall be made only when, and to the extent, specifically authorized by the procuring activity.

3.4.13 Stowed equipment. All on vehicle equipment (OVE) items shall be stowed (combat loaded) on the vehicle in spaces provided.

3.5 Break-in run.

3.5.1 Preparation. Prior to break-in run, the vehicle shall be completely assembled and all adjustments completed. The vehicle, including all components requiring lubrication shall be lubricated as specified in production lubrication chart shown on EPL 10941625. The vehicle shall be supplied with fuel conforming to Grade DF 2 of VV-F-800.

3.5.2 Oil pressure and temperature. Prior to the start of the break-in run, the engine shall be operated at 750 to 1000 revolutions per minute (rpm) until lubricating oil is at operating pressure and temperature. If at any time during the break-in run the engine has been stopped for at least 30 minutes, the engine shall again be operated, as above, before continuing break-in run. Proper oil pressure and temperature shall be maintained during operation on level ground, 60 percent grades and 40 percent side slope.

3.5.3 Distance operated. Prior to tests, each vehicle shall be given a break-in run on smooth, level, hard-surfaced roads, except a section of the road shall be as shown in figure 1. The vehicle shall be operated as specified in table II and over the section of road shown in figure 1, in "a" and "b" division of the break-in run in both directions for the purpose of observing the function and performance of various parts, assemblies controls, and suspension system.

Table II. Speeds and Distance for Break-in Run

Division of run	Speed M.P.H	Distance Miles
a	0 to 10	10
b	11 to 20	25
c	21 to 30	5

3.5.4 Reverse operation. After each division of the break-in run, vehicle shall be stopped, the engine allowed to idle for not less than 2 minutes, and the vehicle driven in reverse for a distance of not less than 50 feet or simulated operation in skid-type dynamometer at speeds between 5 and 15 mph.

3.5.5 Condition after run. After completion of the break-in run, or simulated break-in run, the engine shall idle smoothly. Prior to vehicle being submitted to the Government for acceptance, all suspension wheel bearings shall be checked to determine if adjustments are correct and readjusted as required. There shall be no defects indicative of damaged components, or maladjustments that may cause faulty vehicle operation.

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3.6 Performance. A complete vehicle, combat loaded or with a simulated load of equal weight (see 3.4.13) and serviced, shall perform as specified herein. Ammunition may be simulated in size and weight. A simulated load equal in weight of the gun shall be added. Vehicle, serviced (see 3.5.1) and equipped for existing climatic conditions, shall operate as specified without special equipment. Unless otherwise specified, performance shall be demonstrated on smooth, level, hard-surfaced roads free of loose material, except a section of road shall be as shown in figure 1.

3.6.1 Environmental. The vehicle shall operate in ambient air temperature of plus 115° to minus 25°F, and with special equipment (see 3.4.2) installed at temperature range of minus 25°F to minus 65°F.

3.6.2 Power package and train. When power package is installed in accordance with applicable drawings, it shall function throughout all gear and speed ranges without abnormal noises, loss of lubricants, and damage that my cause failure of vehicle. The power train and associated controls shall be capable of operation throughout all speed and steering ranges without binding of linkages, objectionable noises, loss of lubricants, grabbing, and chattering or slippage when controls are applied to control, stop, and hold vehicle.

3.6.3 Air cleaner. With a maximum air flow of 600 cubic feet per minute, the air cleaner shall be capable of removing not less than 99.8 percent of dust and other foreign material from the engine intake air when tested in accordance with MIL-A-62048.

3.6.4 Water removal (bilge pump). The bilge pump installed in accordance with applicable drawings shall remove water from the vehicle at a minimum rate of 40 gallons per minute (gpm), with a discharge head of 4 1/2 feet and with pump motor operating on 27.5 volts impressed at the motor.

3.6.5 Speeds.

3.6.5.1 Speeds no load. Vehicles shall be capable of operating at a Sustained speed of 30 mph and shall be capable of maintaining a minimum of 2.5 mph in lowest gear without towed load. The vehicle shall be capable of obtaining a maximum speed of 34 mph. With vehicle traveling between 25 mph and maximum speed, the drift from a straight line shall be not more than 3 feet in 100 feet.

3.6.5.2 Speeds with towed load. The vehicle shall be capable of towing a track laying free rolling vehicle of equal weight at a sustained speed of 15 mph.

3.6.6 Acceleration. The vehicle on dry, smooth, level, hard surfaced roads free of loose material shall from a standing start be capable of accelerating through a distance of 300 feet in not more than 15 seconds.

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3.6.7 Climbing. Vehicle shall climb a prepared 60 percent longitudinal grade in forward and reverse gear from a standing start on the grade.

3.6.8 Slope operation. Vehicle shall be capable of sustained automotive operation on 40 percent side slopes with each side of vehicle up slope.

3.6.8.1 Engine starting on grades and side slopes. When standing on a 60 percent longitudinal grade, for not less than 2 minutes, with engine operating under no load between 750 and 1000 rpm, the engine shall be stopped for not less than two minutes. The engine shall restart in not more than one minute when headed up and headed down grade. A similar engine starting test shall be conducted on 40 percent side slope with each side of vehicle up slope.

3.6.9 Braking.

3.6.9.1 Stopping. Vehicle traveling at 20 mph, on dry, smooth, level, hard-surfaced roads free of loose material shall be capable of being stopped by service brake within a distance of 35 feet from point of brake application. Vehicle drift during stopping shall not exceed three feet.

3.6.9.2 Holding. Vehicle, standing on a 60 percent longitudinal grade with brakes applied and locked for parking, transmission in neutral, vehicle shall remain stationary when headed up and down grade. For this operation, vehicle shall be combat loaded or with a simulated load of equal weight.

3.6.10 Turning. The vehicle shall be capable of making continuous 360 degree turns to the right and to the left. (With the vehicle in gear steer the rear point of outer edge of vehicle shall be within a 100 foot diameter circle at all times.) When clutch brake steer is employed, the vehicle's track pattern shall be within a 35 foot diameter circle at all times.

3.6.11 Fording shallow water and floatation operation.

3.6.11.1 Fording shallow water. Vehicle shall ford hard-bottom salt or freshwater crossing up to 42 inches in depth, without special equipment. With vehicle operating in water up to 42 inches in depth for 15 minutes, the accumulation of water shall be not more than 1 inch in the bottom of the hull.

3.6.12 Trench crossing. Vehicle shall cross trenches 36 inches in depth and 72 inches in width at low speeds without stalling or damage to vehicle.

3.6.13 Vertical obstacle. Vehicle shall cross vertical obstacles up to 21 inches in height at low speeds without stalling or damage to vehicle.

3.6.14 Cab and weapon system controls (non-combat loaded).

3.6.14.1 Cab control system.

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3.6.14.1.1 Cab ring friction. With the vehicle horizontal within 2 degrees the turret assembly mounted to the hull, and the turret complete with slip rings, but without traverse gear box output pinions installed, the torque required to rotate the turret shall not exceed 700 foot pounds after breakaway.

3.6.14.1.2 Cab traversing speeds. The cab shall be controllable in traverse at all speeds between 0.5 mils per second and 3 rpm. The tracking speed shall range from not more then 0.5 mils per second to not less than 65.2 mils per second and the transition between these two speeds shall be smooth and gradual. At speeds above 65.2 mils per second, cab speed shall increase with increased control handle displacement until maximum velocity is reached. The cab speed at this point shall be not less than 3 rpm. The speed shall remain stable at all times. These requirements are applicable to gunner power controller only.

3.6.14.1.3 Manual traverse effort. With the vehicle horizontal within 2 degrees, the following cab performance is required. The average force applied tangentially to the traverse hand-crank at a uniform rate, to maintain cab movement, shall not exceed 16.5 pounds with 10 mils per turn rate. With 5 mils per turn rate, effort to maintain cab movement shall not exceed 9 pounds.

3.6.14.1.4 Manual traverse response rate. With traverse mechanism shifted to high range, one revolution of the hand traverse crank shall rotate the cab 9 mils minimum. With traverse mechanism shifted to low range, one revolution of the hand traverse crank shall rotate the cab 5 mils minimum.

3.6.14.1.5 Weapon slope operation. With the vehicle pitched up to 6 degrees and canted up to 5 degrees, the control system shall be capable of rotating the turret 360 degrees right and left with bath power and manual controls.

3.6.14.1.6 Hydraulic pressure limit switch. When power traversing the cab at a rate not to exceed 65 mils per second, the hydraulic system pressure switch shall actuate the hydraulic control power pack motor at a control system pressure of 880 to 970 pressure per square inch (psi), to be indicated on the pressure gage. With the gunner's and number one man power and manual controls in neutral position, the hydraulic pressure switch shall deactivate the hydraulic power pack motor at a control system pressure as indicated on the gage between 1150 and 1500 psi.

3.6.14.2 Weapon control system.

3.6.14.2.1 Elevation system protection. There shall be no damage to the control system or any component as a result or positioning the gun mount against the elevation and depression stops by power or manual controls. Power and manual controls shall be regained instantaneously from this action. The gun mount shall engage the weapon stops at elevation of between minus 36 mils to minus 54 mils and between plus 1315 mils and plus 1333 mils.

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3.6.14.2.2 Elevation and depression limits. The gun mount shall be capable of movement throughout the range of minus 2 degrees to minus 3 degrees to plus 74 degrees in elevation for all directions of cab traverse throughout 360 degrees.

3.6.14.2.3 Control transfer. Control shall be selective at the gunner's position and shall be instantaneously operative when selector switch is actuated. Gunner shall have power control of the traverse axis and the No. 1 man shall have power control of the elevation axis. When selected, the traverse control can be transferred to the No. 1 man, thus giving him control of both traverse and elevation. With controls in centered position, there shall be no movement of the elevation cylinder rod or cab caused by turning the cab control switch on or off.

3.6.14.2.4 Control system deadspot. The cab and gun control system deadspot as measured at the controllers shall not exceed 7 degrees handle movement from neutral center in any direction to initiate elevation cylinder rod and cab movement, for gunner's control; for No. 1 man control, the requirement shall apply to elevation cylinder only. The neutral center shall be defined as that handle position which divides the initial free handle movement (backlash) in half. The deadspot angles each side of the neutral center in the traverse axis must be within 2 degrees, and the deadspot angles in the elevation axis must be equal within 2 degrees.

3.6.14.2.5 Hydraulic control system leakage. With the hydraulic control system fully charged and power control handles in the neutral position, there shall be no evidence of system pressure loss during a ten minutes period (observed at the system pressure gage).

3.6.14.2.6 Hydraulic fluid. When all requirements of the cab control system specified herein have been met and with all hydraulic fluid discharged from the 2 gallon accumulator, the hydraulic fluid in the control power pack reservoir shall contain not more than 0.1 percent water. The hydraulic fluid in the control power pack reservoir shall be free of all metallic or abrasive particles and paint no larger than 100 microns (.004) in diameter.

3.6.14.2.7 Hydraulic accumulator. When all requirements of the turret and gun control system specified herein have been met and with all hydraulic fluid discharged from the 2 gallon accumulator, the precharge nitrogen gas pressure in the 2 gallon accumulator shall be equal to the initial gas pressure corrected for permissible leakage and temperature change. When measuring the final precharge pressure and with the control handle in neutral position, and the cab and gun control system pressure gage stabilized at a zero pressure setting, the system pressure gage needle shall show no visible movement for a period of three minutes.

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3.6.15 Mount. The gun mount shall conform to MIL-M-45212.

3.7 Radio suppression. Complete vehicle shall be radio interference suppressed in accordance with tactical vehicle requirements of MIL-STD-461.

3.8 Painting, marking and data plates.

3.8.1 Painting. Unless otherwise specified, the exterior and interior of cab and hull, components, assemblies, sub-assemblies, and parts that require painting shall be prepared and painted in accordance with applicable drawings and MIL-STD-193. Vehicle exterior paint and nonskid paint shall match color chip X24087 (olive drab) of FEDERAL STD NO. 595.

3.8.2 Marking. Marking shall be as specified by FRG.

3.8.3 Name, shipping, and service data plates. Data plates shall be photographic aluminum plate (sensitized). The plates shall have the photosensitive silver compounds imbedded within the oxide layer; and after photographic processing, the letters, numbers and all other characters shall be sealed in the oxide layer by suitable chemical treatment of the layer. The plates shall have a black background, with white letters and letter blanks, except that caution and warning plates shall have red letters and letter blanks. The size of the plate, type and size of lettering, and the information to be included shall be in accordance with applicable drawings or as specified by the procuring activity. Thickness of the plates shall be .030 inch minimum. The plates shall meet the performance requirements of MIL-P-514.

3.9 Fire extinguishers.

3.9.1 Fixed. The fixed fire extinguisher system, when installed in accordance with applicable drawings, shall be readily accessible for servicing and operation internally and externally. In ambient air temperatures of 72°F, the maximum effort required to discharge the system, either internally or externally, shall be not more than 55 pounds.

3.9.2 Portable. The portable fire extinguisher, when installed in accordance with applicable drawings, shall be readily accessible for servicing and use.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.1.1 Parts and components. Parts, components and assemblies shall be inspected in accordance with the drawings listed on EPL 10941625, and Supplementary Quality Assurance Provisions Index (LSQAP) 10941625.

4.1.2 Contractor's quality control system. Unless otherwise specified by the procuring activity, the contractor shall provide and maintain an effective Inspection and Quality Control System acceptable to the Government covering the supplies under the contract. A current written description of the system shall be submitted to the contracting officer prior to Initiation of production. The contractor will not be restricted to the inspection station or to the method of inspection listed provided that an equivalent control is included in the approved quality control procedure. The contractor shall notify the Government of and obtain approval for any change to the written procedure that might affect the degree of control required by this specification or other applicable documents referenced herein.

4.1.3 Government verification. All quality assurance operations performed by the contractor will be subject to Government verification at unscheduled intervals. Verification will consist of (a) surveillance of the operations to determine that practices., methods, and procedures of the written inspection plan are being properly applied, and (b) Government product inspection to measure quality of product offered for acceptance. Deviation from the prescribed or agreed-upon procedures or instances of poor practices which might have an effect upon the quality of the product, will immediately be called to the attention of the contractor. Failure of the contractor to promptly correct deficiencies discovered shall be cause for suspension of acceptance until corrective action has been made or until conformance of product to prescribed criteria has been demonstrated.

4.1.4 Qualified products. The contractor's inspection records shall be checked to determine contractor conformance to 3.3.1.

4.2 Preproduction and initial production inspection.

4.2.1 Preproduction inspection. A preproduction vehicle when furnished, shall be inspected to ascertain vehicle conformance with the requirements of their specification. Inspection shall be performed at a location approved by the Government. Failure of the preproduction model to comply with any of the requirements specified shall be cause for refusal by the Government to conduct a retest until corrective measures satisfactory to the Government have been taken.

4.2.2 Initial production inspection.

4.2.2.1 In-process inspection. During fabrication of the initial production vehicle (see 3.2), an in-process inspection will be conducted by USATACOM Product Assurance representatives to evaluate conformance of materials and workmanship to requirements of specified documents. The vehicle, suspension and electrical components shall be available for inspection. These inspections shall be made at the contractor or subcontractor's facility prior to application of primer and paint. Processing and welding procedures, quality system and inspection records will be reviewed and evaluated during this inspection.

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4.2.2.2 Completed first vehicle inspection. The first production vehicle of each model on each contract shall be road tested and completely inspected by the contractor for conformance to the requirements of contract and specifications. The applicable road test shall be in accordance with paragraph 4.5.1.1. Upon completion, the contractor shall submit the vehicle (and make available all inspection records and certifications) to the responsible Government inspection element at contractor's plant for preliminary inspection.

4.2.2.2.1 Preliminary inspection. The vehicle shall receive a preliminary inspection by the responsible Government inspection element. The preliminary inspection shall be complete except for road test requirements of paragraph 4.5.1.1 which will be conducted jointly during provisional inspection to be accomplished by representatives of TACOM Product Assurance Directorate as specified in paragraph 4.2.2.2.2.

4.2.2.2.2 Provisional inspection. The first complete production vehicle shall be subject to provisional inspection at the contractor's plant by representatives of TACOM Product Assurance Directorate and shall further provide any other inspection assistance as may be required. At the time of this inspection, the contractor shall make available his inspection plan, inspection records and certifications pertinent to the vehicle and components.

4.2.2.2.3 Corrections. Corrections found necessary as a result of above inspections shall be made by the contractor.

4.2.2.2.4 Vehicle disposition. After completion of inspection, the first production vehicle of a specific model shall remain at the manufacturing facility, as a production sample, and be the last vehicle shipped on the contract. The vehicle may be released at the discretion of the contracting officer. The contractor shall service and maintain the vehicle during this period in accordance with applicable documents for care and preservation while in storage.

4.2.2.3 Final approval and acceptance. Final approval and acceptance by the Government of the first vehicle of a specific model shall be withheld until the 4000 mile test of the second vehicle produced has been completed and a final determination has been made regarding conformity of the vehicle to contractual requirements, including, but not limited to workmanship and materials.

4.2.3 Initial production test. After completion of preliminary inspection of the first vehicle, a second vehicle selected from first month's production shall be subjected to examination and tests specified herein, and the 4000 mile test specified in table III. These tests shall be performed by the Government at a site approved by the Government, and will require a maximum of 90 days with a target completion date of 60 days. Under separate contractual authority, the contractor will expeditiously furnish repair parts, as required to adequately support the above test. Delays caused by vehicle breakdown and failure of the contractor to adequately support the vehicle with parts during the tests shall not be the responsibility of the Government.

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Table III. 4000 Mile Teat Combat Loaded or Simulated Load
(Government Proving Ground)

Course	Mileage & Speeds
Hard surface roads	1000 miles at varying speeds up to maximum
Gravel and dirt roads	2000 miles at varying speeds up to maximum with 300 miles of this distance over hilly conditions
Level cross-country	600 miles at varying speeds up to maximum
Hilly cross-country	400 miles at varying speeds up to maximum

4.2.3.1 Reliability (automotive) verification. To determine conformance to 3.2.1, reliability requirement will be verified as a point estimate while the vehicles are subjected to Initial production test. The mean miles between failure (MMBF) shall not be less than specified.

4.2.3.2 Durability (automotive) verification. To determine conformance to 3.2.2, the durability requirement will be verified as a point estimate while the vehicle is subjected to initial production test.

4.2.3.3 Maintainability (Automotive) verification. To determine conformance to 3.2.3, the maintainability requirements will be verified during initial production test.

4.2.3.4 Test failure. Failure of the vehicle as the result of any deficiency of a workmanship or materials nature during, or as a result of, the 4000 mile test, shall be cause for rejection of the vehicle. Further, the Government may refuse to continue acceptance of production vehicles until evidence has been provided by the manufacturer that corrective action has been taken to eliminate the deficiency. Any deficiency found during, or as a result of the 4000 mile test shall be prime facie evidence that all vehicles already accepted prior to completion of the 4000 mile test are similarly deficient unless evidence satisfactory to the contracting officer is furnished by the contractor that they are not similarly deficient. Such deficiencies on all vehicles shall be corrected by the contractor at no cost to the Government.

4.3 Break-in run, (track). To determine conformance to 3.5.1, 3.5.2, and 3.5.3 each vehicle shall be lubricated and serviced prior to the break-in run. The run shall be on smooth, level, hard-surface roads, and for the distance specified in table II. The vehicle shall be operated in reverse after each division as specified in 3.5.4. The undulating portion of the test shall be accomplished twice in each direction at speed between 5 and 10 mph during division "a" and between 11 and 14.5 mph during division "b".

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4.3.1 Simulated break-in run. To determine conformance to 3.5.1, 3.5.2, and 3.5.3, a simulated break-in run (see table IV) may be accomplished, in the Army Skid Type Dynamometer, in lieu of the run specified in 4.3. Prior to the break-in run, each vehicle shall be lubricated, serviced and warmed up for not less than twenty minutes. Records shall be kept of the vehicle instrument readings. The final ten miles of the total break-in run shall be on smooth, level, hard surface roads. During this ten mile test, the vehicle shall be subjected to undulation operation, in accordance with 3.5.3, by being operated over the section of road conforming to figure I. After completion of divisions b and d, table IV, the vehicle shall be operated in reverse as specified in 3.5.4. The undulating portion of the test shall be accomplished twice in each direction at speeds between 5 and 10 mph.

Table IV. Dynamometer and Test Track

Division of Run	Speed mph	Miles of test	Test conditions
a	0 to 10	10	Dynamometer
b	11 to 20	15	Dynamometer
c	Maximum	5	Dynamometer
d	10 to Max	10	Track (road)

4.3.2 Condition after run. After the break-in run, the vehicle shall be inspected to determine conformance to 3.5.5.

4.4 Inspection provisions.

4.4.1 Lot size. An inspection lot shall consist of all vehicles of one type, from an identifiable production period, submitted at one time for acceptance inspection.

4.4.1.1 Sampling for inspection. For the purpose of visual, dimensional, and primary functional inspection, a representative sample shall be selected, from each inspection lot in accordance with inspection level II of MIL-STD-105.

4.4.2 Examinations. Visual, dimensional, and primary functional examination shall consist of examination of the vehicle for conformance to the applicable drawings and this specification. Examination shall be performed against the classification of defects and with the acceptable quality levels shown in table V. Defects found in Government furnished equipment, unless caused by vehicle contractor, will be classified a special and shall not be used for lot rejection criteria. The following constitutes a part of the classification of defects (see 6.4):

(a) Any drip constitutes a major defect when the vehicle has been standing idle and the components are at ambient temperature.
(b) Any leaks that occurs at a static fit, metal to metal or gasket combination shall constitute a major defect.

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Table V. Classification of Defects
Major Defects AQL 25 Defects/100 Units

	Characteristics	Defects	Method of Inspection
101	Engine	Malfunction, unusual noise, leaks mounting improper damage	Visual & Functional
102	Engine governors	Improper setting, malfunction	Functional
103	Power train	Malfunction, unusual noise, mounting, improper damage	Visual & Functional
104	Fuel system components	Malfunction, damage, leaks	Visual & Functional
105	Fuel & oil lines valves & hose	Defective, clearance improper malfunction, leaks	Visual & Functional
106	Cooling system components (engine and transmission)	Malfunction, damage, leaks, unusual noise	Visual & Functional
107	Exhaust system components	Damage, leaks	Visual
108	Electrical system components (includes wiring & instrumentation)	Malfunction, damage, unusual noise	Visual & Functional
l09	Suspension system components	Damage, tire or track pad separation improper clearance or alignment welding defects, leaks, malfunction	Visual & Functional
110	Hull	Structure and welding defect	Visual
111	Control and adjusting mechanisms	Malfunction, improper clearance	Visual & Functional
112	Doors, fenders, grilles, panels, ducts, stowage brackets, boxes, racks and stops	Malfunction, torsion spring and hold-open devices	Visual & Functional
113	Drain valves	Valve seating improper	Visual
114	Cushions, seats, rests & crash pads	Damage, seat control malfunction	Visual & Functional

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Table V. Classification of Defects (Cont'd)
Major Defects AQL 25 Defects/100 Units

	Characteristics	Defects	Method of Inspection
115	Howitzer travel lock	Locking lug, malfunction, damage	Visual & functional
116	Traversing and elevating mechanisms	Clearance improper, component damage	Visual & functional
117	Recoil mechanisms replenisher & lines	Component damage, indicators of lines defective	Visual & functional
118	Weapon control system (includes panels, solenoids, switches wiring & other components)	Component damage, defective wiring	Visual & functional
119	Misc. items, accessories	Malfunction, damage, missing	Visual & functional
120	Fire extinguisher system components	Seals missing or defective	Visual
121	Speed	Nonconformance	Functional
122	Acceleration	Nonconformance	Functional
123	Turning ability and steering	Nonconformance, wander, side-pull, unusual noise	Functional
124	Braking ability (parking & service brakes)	Nonconformance: grade holding or stopping distance, equalization improper, unusual noise	Functional
125	Fording (shallow)	Water leakage excessive	Visual

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Table V. Classification of Defects
Minor Defects AQL 150 Defects/100 Units

	Characteristics	Defects	Method of Inspection
201	Engine	Improper component assembly or installation	Visual & Functional
202	Engine governors	Missing or defective seals	Functional
203	Power train	Improper component assembly or installation	Visual & Functional
204	Fuel system components	Improper assembly, installation or clearance	Visual & Functional
205	Fuel & oil lines valves & hose	Improper assembly or installation	Visual & Functional
206	Cooling system components (engine and transmission)	Improper assembly or installation	Visual & Functional
207	Exhaust system components	Improper assembly, installation or clearance	Visual
208	Electrical system components (includes wiring & instrumentation)	Improper assembly, or installation	Visual & Functional
209	Suspension system components	Improper assembly, installation, bolt or nut torque and track tension	Visual & Functional
210	Control and adjusting mechanisms	Improper assembly or installation, welding defects	Visual & Functional
211	Doors, fenders, grilles, panels, ducts, stowage brackets, boxes, racks and stops	Improper fit, adjustment, assembly or installation	Visual & Functional

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Table V. Classification of Defects (Cont'd)
Minor Defects AQL 150 Defects/100 Units

	Characteristics	Defects	Method of Inspection
212	Drain valves	Improper assembly installation, or control adjustment	Visual
213	Cushions, Beats rests & crash pads	Improper fit, installation or adhesion	Visual & functional
214	Howitzer travel lock	Improper assembly or installation	Visual & functional
215	Traversing and elevating mechanisms	Improper assembly or installation	Visual & functional
216	Recoil mechanisms replenisher & lines	Improper assembly, installation, fluid level and type improper	Visual & functional
217	Weapon control system (includes panels, solenoids, switches wiring & other components)	Improper clearance, assembly, installation, circuit identification, protection or markings	Visual & functional
218	Misc. items, accessories	Improper assembly or installation	Visual & functional
219	Fire extinguisher system components	Data card incomplete, improper assembly or installation	Visual
220	Paint, protective & anti-skid coatings	Application, coverage and materials improper	Visual
221	Lube fittings & lubrication	Fittings missing, defective or improperly installed, lubrication improper	Visual
222	Decals, marking data instruction plates	Incomplete, data, missing, improper location or size, illegible	Visual

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4.4.2.1 Unclassified defects. All defects having no bearing on function, safety, interchangeability or life, but are considered departures from good workmanship will be noted in writing. Workmanship deficiencies falling within this category and recurring in 5 consecutive lots or ten lots or more within a thirty-day period will be added to the minor classification with no increase in AQL's.

4.5 Classification of tests.

(a) Acceptance tests, 4.5.1.1 through 4.5.1.22.6.
(b) Control tests, 4.5.2 through 4.5.2.9.
(a) Comparison test (CT), 4.5.3.2 through 4.5.3.14.

4.5.1 Tests.

4.5.1.1 Acceptance tests. To determine conformance to 3.6, after the break-in run (see 3.5), each vehicle shall be operated for a distance of eight miles or more and shall be inspected as specified in 4.4.2. These acceptance tests shall include all acceptance tests specified at the place of manufacture in table VI. Vehicles shall be completely assembled and serviced, with fuel, lubricants, and fluid as specified, but need not be combat loaded, except holding and flotation tests as specified in 4.5.1.18 and 4.5.1.20. For these tests, vehicles shall be operated on roads as specified. The acceptance tests for the automotive and cab control and gun control system may be ran simultaneously where applicable. After the tests, the vehicle shall be examined for evidence of fuel lubricant leakage, or other deficiencies.

4.5.1.1.1 Test failure. If a vehicle fails to pass any acceptance test specified herein, the Government inspector shall stop acceptance of subsequent lots until evidence has been provided by the contractor that the corrective action has been taken.

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Table VI. Location for Tests

Title	Paragraph Number	Place of Manufacture	Government Proving Ground
Acceptance tests	4.5.1.1	X	X
Hatch seal check	4.5.1.2	X	X
Controls and linkage check	4.5.1.3	X	X
Adjustment mechanism check	4.5.1.4	X	X
Fastener and lock check	4.5.1.5	X	X
Fuel system leakage check	4.5.1.6	X	X
Hydraulic reservoir and line check	4.5.1.7	X	X
Light check	4.5.1.8	X	X
Bilge pump electrical check	4.5.1.9	X	X
Air cleaner blower electrical check	4.5.1.10	X	X
Personnel heater electrical check	4.5.1.11	X	X
Communication equipment and slip ring electrical check	4.5.1.12	X	X
Power package and train test	4.5.1.13	X	X
Speed test	4.5.1.14	X	X
Acceleration test	4.5.1.15	X	X
Engine starting on grades and side slope tests	4.5.1.16	X	X
Stopping tests	4.5.1.17	X	X
Holding test	4.5.1.18	X	X
Turning test	4.5.1.19	X	X
Shallow water fording test	4.5.1.20	X	X
Cab traversing speed test	4.5.1.21.1	X	X
Manual traverse effort check	4.5.1.21.2	X	X
Manual traverse response rate check	4.5.1.21.3	X	X
Weapon slope operation check	4.5.1.21.4	X	X
Hydraulic pressure limits switch test	4.5.1.21.5	X	X
Evaluation system protection check	4.5.1.22.1	X	X
Evaluation and depression limits and time check	4.5.1.22.2	X	X
Control transfer check	4.5.1.22.3	X	X
Control system deadspot check	4.5.1.22.4	X	X
Hydraulic control system leakage check	4.5.1.22.5	X	X
Hydraulic accumulator test	4.5.1.22.6	X	X
Control tests	4.5.2	X	X
Frequency	4.5.2.1	X	X
Contamination test	4.5.2.3	X	X
Fuel system slope test	4.5.2.4	X	X
Fuel tank and line test	4.5.2.5	X	X
Stowed equipment test	4.5.2.6	X	X
Cab ring friction test	4.5.2.7	X	X
Hydraulic fluid test	4.5.2.8	X	X
Fire extinguisher test	4.5.2.9	X	X

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Table VI. - Location for Tests (Cont'd)

Title	Paragraph Number	Place of Manufacture	Government Proving Ground
Comparison tests	4.5.3		X
Winterization equipment test	4.5.3.2		X
Carbon monoxide concentration test	4.5.3.3		X
Cooling system tests	4.5.3.4		X
Communications equipment test	4.5.3.5		X
Temperature operational tests	4.5.3.6		X
Air cleaner operational check	4.5.3.7		X
Water removal (bilge pump) test	4.5.3.8		X
Speeds with towed load test	4.5.3.9		X
Climbing test	4.5.3.10		X
Slope operational test	4.5.3.11		X
Trench crossing test	4.5.3.12		X
Vertical obstacle test	4.5.3.13		X
Radio suppression test	4.5.3.14		X

4.5.1.2 Hatch seal check. To determine conformance to 3.4.1.3, the batch shall be closed and locked, a spray of free flowing water directed on hatch for a period of 3 minutes without evidencing leakage. After 2 minutes, following the 3 minute spray period, the hatch shall be opened and no water shall enter the vehicle. This test may be conducted during shallow water fording.

4.5.1.3 Controls and linkage checks. To determine conformance to 3.4.4, 3.4.5 all controls, and throttle linkage shall be operated and checked for functional requirements.

4.5.1.4 Adjustment mechanism check. To determine conformance to 3.4.6, all adjustment mechanisms shall be checked for proper adjustment and functional requirements.

4.5.1.5 Fastener and lock check. To determine conformance to 3.4.7, all fastener and locking devices shall be operated and checked for positive fastening and locking.

4.5.1.6 Fuel system leakage check. To determine conformance to 3.4.8 through 3.4.8.2, the fuel system shall be pressurized with dry air at a pressure of 3 to 4 pounds per square inch (psi), prior to engine installation, and check fittings, lines, and fuel tank seals for leaks; by applying soapy water or equivalent.

4.5.1.7 Hydraulic reservoir and line check. To determine conformance to 3.4.9, prior to initial operation of the hydraulic system the reservoir and lines shall be checked for leaks and cleanliness.

4.5.1.8 Light check. To determine conformance to 3.4.11.1, all lights shall be operated during and after break-in run throughout all stages of operation, and checked for functional requirements.

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4.5.1.9 Bilge pump electrical check. To determine conformance to 3.4.11.2, the bilge pump electrical controls shall be turned on and the pump checked for operation.

4.5.1.10 Air cleaner blower electrical check. To determine conformance to 3.4.11.3, the air cleaner electrical controls shall be turned on and the blower checked for operation and air flow.

4.5.1.11 Personnel heater electrical check. To determine conformance to 3.4.11.4, the personnel heater electrical controls shall be turned on and the heater checked for operation.

4.5.1.12 Communication equipment and slip ring electrical check. To determine conformance to 3.4.11.5 and 3.4.11.6, the electrical circuits supplying current to the communication equipment and slip ring shall be checked at all points to assure continuity.

4.5.1.13 Power package and train test. To determine conformance to 3.6.2, the vehicle shall be operated throughout all gear and speed ranges, the power plant, power train, final drives, tracks, suspension system, and applicable controls.

4.5.1.14 Speed tests. To determine conformance to 3.6.5.1, the vehicle shall be operated at specified speeds and checked for functional requirements.

4.5.1.15 Acceleration test. To determine conformance to 3.6.6, the vehicle shall be accelerated from a standing start to the required speeds and the time checked.

4.5.1.16 Engine starting on grades and side slope tests. To determine conformance to 3.5.2 and 3.6.8.1, the engine shall start, oil pressure and temperature shall be maintained when operating on specified grades and slopes. The 60 percent grade test may be conducted in conjunction with brake holding test.

4.5.1.17 Stopping test. To determine conformance to 3.6.9.1, the vehicle shall be operated on dry level hard-surfaced road free of loose material, and the results of 3 stopping tests shall be averaged to make this determination.

4.5.1.18 Holding test. To determine conformance to 3.6.9.2, the vehicle shall be driven up a specified grade, brakes applied and vehicle observed for movement, then with brakes locked for parking the vehicle shall be observed for movement. During this test vehicle shall be combat loaded or a simulated load in weight shall be placed on the vehicle, in a position that will not restrict the flow of engine air (intake or exhaust).

4.5.1.19 Turning test. To determine conformance to 3.6.10, vehicle shall be operated and turned to right and left as specified and checked for radius of turns.

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4.5.1.20 Shallow water fording test. To determine conformance to 3.6.11.1, the vehicle shall be driven into water of specified depth, without special equipment. Water accumulation shall not exceed specified depth in hull bottom after 15 minutes of fording operation. The engine shall idle at 1000 rpm for 15 minutes, be stopped, and shall restart in not more than 3 minutes with all accessories functioning satisfactorily during and after fording.

4.5.1.21 Cab control system tests.

4.5.1.21.1 Cab traversing speed test. To determine conformance to 3.6.14.1.2, by use of the gunner and commander's power controls, the cab shall be traversed clockwise and counterclockwise as specified, and the transition, maximum velocity, and stability of cab observed. The accumulator shall be fully charged prior to beginning the maximum speed test. Maximum traverse speed shall be measured after allowing the cab to accelerate approximately 400 mils from stop. The first revolution of the cab after this acceleration shall be accomplished at the specified rate.

4.5.1.21.2 Manual traverse effort check. To determine conformance to 3.6.14.1.3, the specified average force applied tangentially to the traverse handcrank shall not exceed that specified.

4.5.1.21.3 Manual traverse response rate check. To determine conformance to 3.6.14.1.4, with the traverse mechanism in high range, the response shall not exceed that specified, and when in low range the response shall not exceed that specified.

4.5.1.21.4 Weapon slope operation check. To determine conformance to 3.6.14.1.5, the vehicle shall be checked for functional requirements.

4.5.1.21.5 Hydraulic pressure limits switch test. To determine conformance to 3.6.14.1.6, the cab shall be traversed as specified and the system pressure gage observed for specified pressure as the power pack motor cuts in. With power pack motor running and the controls in their neutral position, the power pack motor shall cut at the specified system pressure.

4.5.1.22 Weapon control system tests.

4.5.1.22.1 Elevation system protection check. To determine conformance to 3.6.14.2.1, with the gun mount driven against stops, the control system shall be checked for damage.

4.5.1.22.2 Elevation and depression limits and time check. To determine conformance to 3.6.14.2.2, the gun mount shall be operated throughout elevation and depression ranges and checked for specified limits.

4.5.22.3 Control transfer check. To determine conformance to 3.6.14.2.3, with controls operated as specified, the control shall take control of weapon as specified.

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4.5.1.22.4 Control system deadspot check. To determine conformance to 3.6.14.2.4, with the cab and gun controls operated as specified, the control system deadspot shall not exceed that specified.

4.5.1.22.5 Hydraulic control system leakage check. To determine conformance to 3.6.14.2.5, with the system fully charged, the pressure drop shall be not less than that specified.

4.5.1.22.6 Hydraulic accumulator test. To determine conformance to 3.6.14.2.7, the elevation and traverse power switch "off" and the cab lock engaged, all oil shall be discharged from the main accumulator by actuating the cab power shut off solenoid button and moving the power traverse handle to extreme right or left until the system pressure gage is stabilized at a zero pressure setting. Leakage from initial precharge condition shall be not more than 2 percent in 24 hours and one psi per degree temperature change. With the specified nitrogen pressure on the gas gage, the controls in neutral position, and the oil pressure at zero the test shall be continued by observing the oil gage for a period of 3 minutes with no gage movement.

4.5.2 Control test. Control tests for maintaining control of those requirements whose inspection would be or too long duration to be included under acceptance tests shall be conducted by the manufacturer as specified herein.

4.5.2.1 Frequency. The first vehicle off the production line, or one of the first ten vehicles produced, and thereafter one per month, shall be checked in accordance with the following control test paragraphs.

4.5.2.2 Test failure. If a vehicle fails to pass any control test specified herein, the Government inspector shall stop acceptance of subsequent lots until evidence has been provided by the contractor that corrective action has been taken.

4.5.2.3 Contamination test. To determine conformance to 3.4.1, 3.4.1.1, 3.4.1.2, the vehicle after the water fording and flotation test shall be examined for contamination of lubricants of the transmission, engine, final drives, and road wheels 2, 3, 4, 5, and one of their corresponding area, on each side of the vehicle. Contamination shall not exceed 2 percent.

4.5.2.4 Fuel system slope test. To determine conformance to 3.4.8, with vehicle headed up and down a 60 percent longitudinal slope, and each side up to 40 percent slope vehicle shall be checked for functional requirements.

4.5.2.5 Fuel tank and lines test. To determine conformance to 3.4.8.1 and 3.4.8.2, the fuel tanks shall be filled to capacity at a rate of not less than 50 gallons per minute. The tanks and lines shall be checked for leaks before and after a two mile road test on smooth, level, hard-surfaced roads.

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4.5.2.6 Stowed equipment test. To determine conformance to 3.4.13, all manufacturer and depot installed "On Vehicle Equipment" (OVE) shall be stowed (combat loaded) on the selected vehicle. The equipment shall be removed after the test. The OVE shall be made available to the contractor, and the contractor shall be permitted to unpackage and repackage for domestic shipment only the equipment used for test.

4.5.2.7 Cab ring friction test. To determine conformance to 3.6.14.1.1, one vehicle in 20 shall be selected. The turret assembly shall be rotated throughout the 360 degree rotation, with a tangential force of 12 pounds applied at the end of the gun tube, parallel to plane of rotation. The torque required to maintain rotation shall be measured, and checked for functional requirements.

4.5.2.8 Reservoir hydraulic fluid test. To determine conformance to 3.6.14.2.6, the hydraulic oil in the control power pack shall be tested for water contamination and cleanliness.

4.5.2.9 Fire extinguisher test. To determine conformance to 3.9.1 and 3.9.2, the fire extinguisher system shall be functionally tested, both sides, subsequent tests of one in each fifty testing alternating sides.

4.5.3 Comparison tests. The Government may select vehicles any time during the contract, production period and subject vehicle to tests listed in table VI and table VII to reveal deficiencies of manufacturer or design that may reduce the effective operation of these items in the field and to compare existing quality with previous standards. These tests shall be conducted at Government laboratories or proving grounds designated by the contracting officer. Selection of vehicles shall be on a spot check basis. Comparison test vehicles shall be combat loaded with all on-vehicle equipment stowed in their applicable spaces to conform to 3.4.13. Vehicle selected shall not include any vehicle previously tested for conformance to 4.5.2 (control tests).

4.5.3.1 Test failure. Failure of any vehicle tested as specified in 4.5.3 to comply with any of the requirements specified in the contract, or any major deficiency of a workmanship or materials nature occurring during, or as a result of the test, may be cause for refusal to continue acceptance of vehicles by the Government until evidence has been provided by the manufacturer that corrective action has been taken to eliminate the conditions.

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Table VII. 2000 Mile Test Combat Loaded or a Simulated Load (Government Proving Grounds)

Course	Mileage & Speeds
Hard surface roads	200 mile at varying speeds up to maximum
Gravel and dirt roads	500 miles at varying speeds up to maximum wit miles of this distance on hilly conditions
Level cross country	650 miles at varying speeds up to maximum
Hilly cross country	650 miles at varying speeds, up to maximum

4.5.3.2 Winterization equipment test. Winterization equipment shall be installed in accordance with applicable drawings and specifications. The vehicle properly serviced and equipped, shall be stabilized and then subjected to respective ambient temperature of plus 25° to minus 65°F. The engine shall satisfactorily start and operate and the personnel heaters shall function as specified in 3.4.2.1.

4.5.3.3 Carbon monoxide concentration test. To determine conformance to 3.4.3, the crew compartment shall be checked for carbon monoxide concentration.

4.5.3.4 Cooling system tests. To determine conformance to 3.4.10, the vehicle shall be operated as specified and cooling systems checked for functional requirements.

4.5.3.5 Communications equipment test. To determine conformance to 3.4.11.5, communication system equipment shall be installed, subjected to tests and checked for functional requirements.

4.5.3.6 Temperature operational tests. To determine conformance to 3.6.1, the vehicle, properly serviced and equipped, shall be subjected to the applicable tests in high and low temperatures.

4.5.3.7 Air cleaner operational check. To determine conformance to 3.6.3, during and after operation, the air cleaners shall be checked for functional requirements.

4.5.3.8 Water removal (bilge pump) test. To determine conformance to 3.6.4, the vehicle shall have sufficient water in hull to operate bilge pump and check the rate of water removal with the specified discharge head.

4.5.3.9 Speeds with towed load test. To determine conformance to 3.6.5.2, the vehicle shall be operated towing specified vehicle, at indicated speeds for a distance of not less than that required to meet performance requirements.

4.5.3.10 Climbing operational test. To determine conformance to 3.6.7, each vehicle selected shall be driven in forward and reverse gear up approved 60 percent grade.

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4.5.3.11 Slope operational test. To determine conformance to 3.6.8, the vehicle shall be operated as specified, on applicable slopes and checked for performance requirements.

4.5.3.12 Trench crossing tests. To determine conformance to 3.6.12, the vehicle selected shall be driven without stalling over trenches as specified and then examined for damage.

4.5.3.13 Vertical obstacle tests. To determine conformance to 3.6.13, the vehicle selected shall be driven without stalling over obstacles as specified and then examined for damage as specified.

4.5.3.14 Radio suppression test. To determine conformance to 3.7, the vehicle shall be subjected to a radio interference suppression test.

5. PREPARATION FOR DELIVERY

5.1 Preparation for delivery and storage shall be as specified by procuring activity.

6. NOTES

6.1 Intended use. The vehicle is intended for use by the Federal Republic of Germany as a combat vehicle.

6.2 Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Preproduction model if required (see 3.1).
(c) Special kits, if required (see 3.4.2).

6.3 Safety precautions. Caution should be exercised in handling carbon dioxide fire-extinguisher cylinders. They should not be dropped, permitted to strike each other, or be handled roughly. Extreme care should be exercised during reinstallation to avoid tripping fire-extinguisher control system since physical injury is highly probable.

6.4 The following shall be used with the classification of defects:

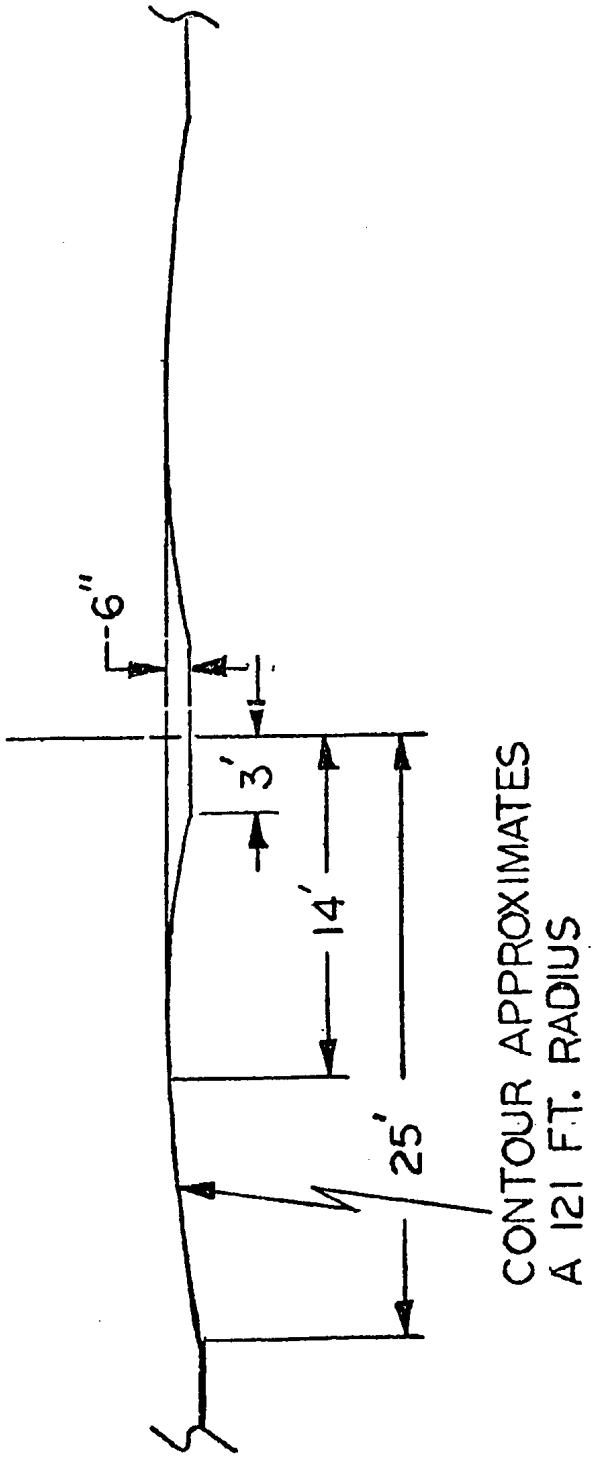
Weep: Any evidence of fluid beyond the seal.
Seep: Any evidence of fluid beyond the seal that does not result in formation of a droplet.
Leak: Any evidence of fluid beyond the seal that results in the formation of a droplet.
Drip: Any evidence of fluid beyond the seal where droplet forms and falls.

Custodian:
Army - AT

Preparing activity:
Army - AT

Project No. 2350-A207

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T01:22Z
Check the source to verify that this is the current version before use.



PROFILE OF TEST TRACK
(ARTIFICIAL BUMP)

FIG. 1

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T01:22Z
Check the source to verify that this is the current version before use.